

FilamentTracker Live Spool Calibration Guide

Version 1.0

Overview

This guide explains how to calibrate the integrated load cell using the built-in Calibration Wizard. Calibration only needs to be performed during initial assembly or after changing the load cell or amplifier.

Before You Begin

- Place the Live Spool on a solid, level surface.
- Ensure the load platform is completely empty.
- Use a known calibration weight (250 g, 500 g, or 1000 g recommended).
- Do not touch the platform during measurements.
- Allow the reading to stabilize before continuing.

Calibration Wizard Steps

1. Welcome

Launch the Calibration Wizard from Settings → Scale → Calibration.

2. Remove Spool

Verify the platform is empty before proceeding.

3. Tare

Press Tare to record the zero offset with nothing on the platform.

4. Place Known Weight

Place the selected calibration weight on the center of the platform.

5. Select Weight

Choose 250 g, 500 g, 1000 g, or enter a custom value.

6. Verification

The wizard compares the measured value to the expected weight. If the difference is acceptable, calibration passes.

7. Finish

Save the calibration. The values are stored and automatically used after reboot.

Verification

After calibration, place your known weight back on the platform and verify the displayed value is accurate. If the reading is inconsistent, repeat the wizard and ensure the weight is centered and the platform is undisturbed.

Troubleshooting

- Reading drifts: Verify the platform is not touching the enclosure and no cables are applying force.
- Calibration fails: Confirm the correct known weight was selected.
- Unstable readings: Check load cell wiring and tighten all screw terminals.
- No weight detected: Verify the NAU7802 (0x2A) is detected and functioning.
- Changing the load cell or amplifier requires recalibration.

Tip: A 500 g calibration weight provides an excellent balance between accuracy and convenience for most users.